Lab 10.1 - Darkness of Recursive World

General Information and Submission Rules:

- By the end of the labwork, you must submit it to Yulearn as **studentID_fullname_labwork8.c**.
- Incorrect naming or file format will result in <u>a loss of 5 points</u> from your total grade.
- Only use concepts covered in the labs so far.

Question 1 - Display Message (10p)

- Create a new function (**welcome**) that displays a welcome message and a list of all available operations to the user.
- This function will expect a decision from a user so that it will execute related functions based on this choice <u>until the user exits</u>.
- Make sure to print each line separately.

```
Welcome to Labwork 10 Darkness of Recursive World! By using this machine you can
do:
1. The Labyrinth of the Hero
2. Shadow Binding
3. Exit
Make your choice:
```

Question 2 - The Labyrinth of the Hero(50p)

- A hero is trapped in a 2D matrix (the labyrinth). Some cells are <u>walls</u>, and one is the <u>exit</u>.
- The hero can only move Up, Down, Left, or Right.
- Create a <u>recursive function</u> (**find_exit**) that takes a matrix (that represents labyrinth).
- The matrix consist of these numbers each has special meaning:

  0: Open path
  1: Wall

9: The exit
2: Visited (to avoid infinite loops)

- In the end, the function should return 1 if the exit is found and 0 if the current path leads to a dead end.
- You can use given matrix as labyrinth:

```
maze[4][4] = { {0, 1, 0, 0},
               {0, 1, 0, 1},
               {0, 0, 0, 1},
               {1, 1, 0, 9} }

Output Example:
>> There is a path!
```

Question 3 - Shadow Binding (40p)

- In a forgotten corner of a cursed library, there exists a scroll protected by Shadow Binding.
- This dark magic hides the truth within a clutter of meaningless symbols.
- To read the secret message, one must cast a recursive spell that slowly dissolves the outer layers of the string, one character at a time, until only the truth - the part wrapped in parentheses - remains.
- Ex. “-*xyz(magic)123”
  (magic) - is the secret
  Ex. empty()space
  () - is the secret
- Create a recursive function (**shadow_binding**) that takes the string that keeps the string. You can define your strings in main.
- In the end, print the secret message.

```
In main:
char secret1[] = "-*xyz(magic)123*-";
char secret2[] = "empty()space";

Output:
>> (magic)
>> ()
```